

Assignment 1

<https://github.com/maxb6299/cis4930-assignment-01>

Setup and Data Collection



```
+ Code + Markdown Colab Run All Restart Clear All Outputs Jupyter Variables Outline ... cis4930 (Python 3.11.16)
```

```
1 print("Hello world!")
```

[1] ✓ 0.0s Python

... Hello world!

```
1 import pandas, numpy, matplotlib, seaborn, scipy, networkx, plotly, requests, tqdm
2 print("All libraries imported successfully.")
```

[7] ✓ 0.0s Python

... All libraries imported successfully.

Downloaded 26603 works in data collection stage:

```
... Fetching works: 26603works [03:34, 124.16works/s]
    Saved 26603 records to fsu_works_2021_2026.jsonl
```

Ran into issues with the “select” param of the API, but once formatted correctly it worked smoothly.

Data Cleaning

The attributes I kept were: id, doi, title, publication year/date, type, venue, citations, authors, primary and all fields. I figured this was more than needed for the following visualizations but it comes at a very low computational and storage cost, so I figured there was little harm in keeping it.

Works with missing concepts, venues, or publication dates simply had an empty cell in the .csv. I figured that although this is important data, it was not absolutely necessary for later visualizations.

Statistics and Charts

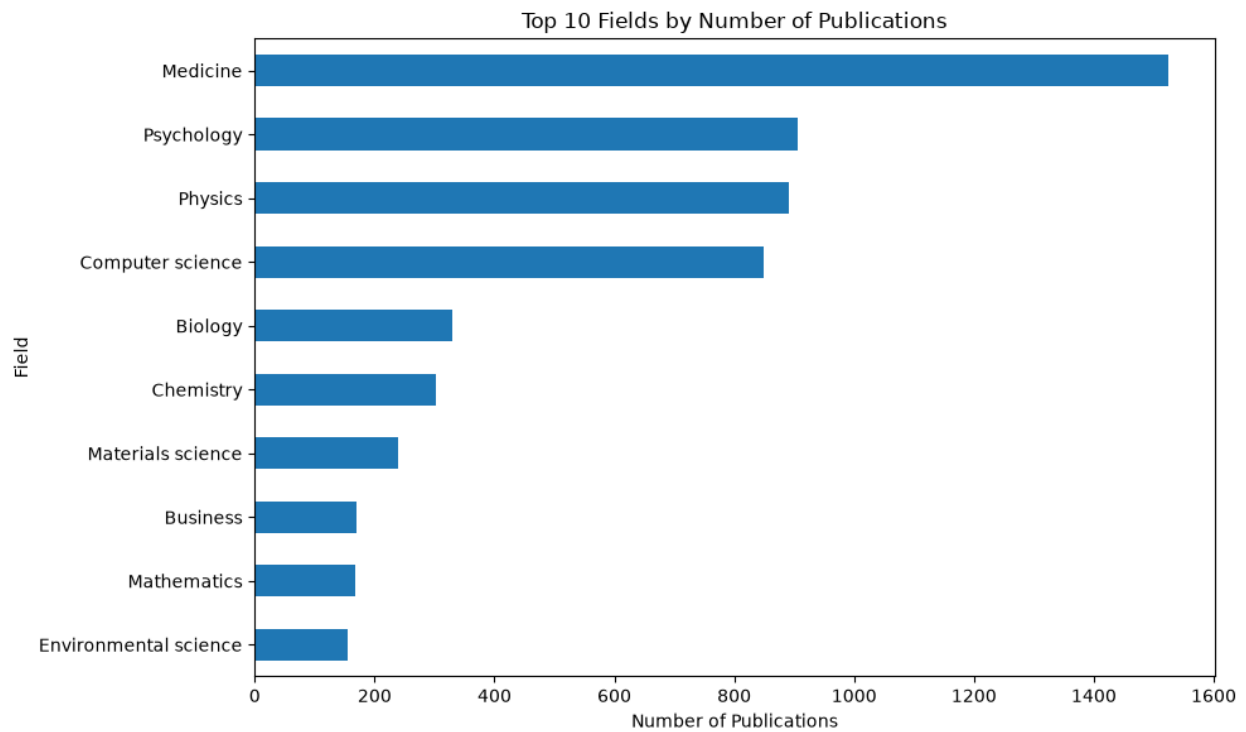


Chart 1 shows FSU's research output is primarily medicine and psychology, however, it hides interdisciplinary papers that involved multiple fields.

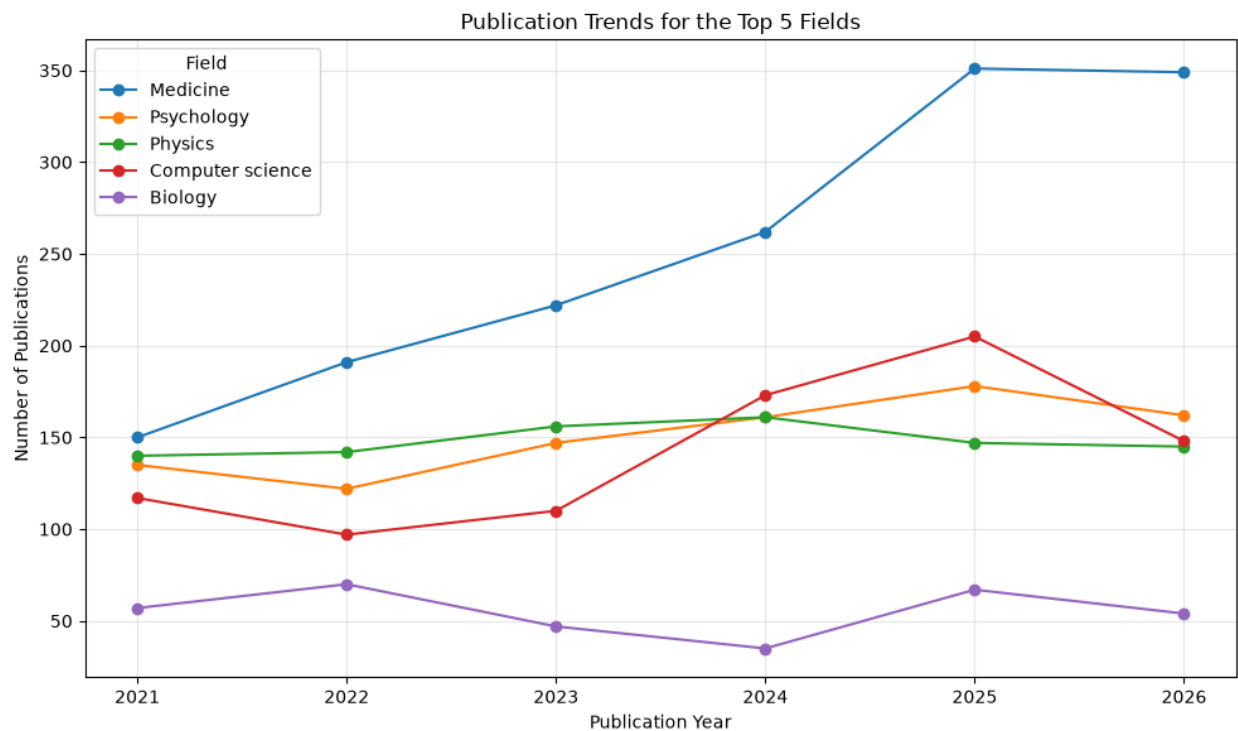


Chart 2 shows FSU's top 5 fields have remained stable or expanded since 2021. However, it hides that the slight decline in 2026 might be that the year is not over yet.

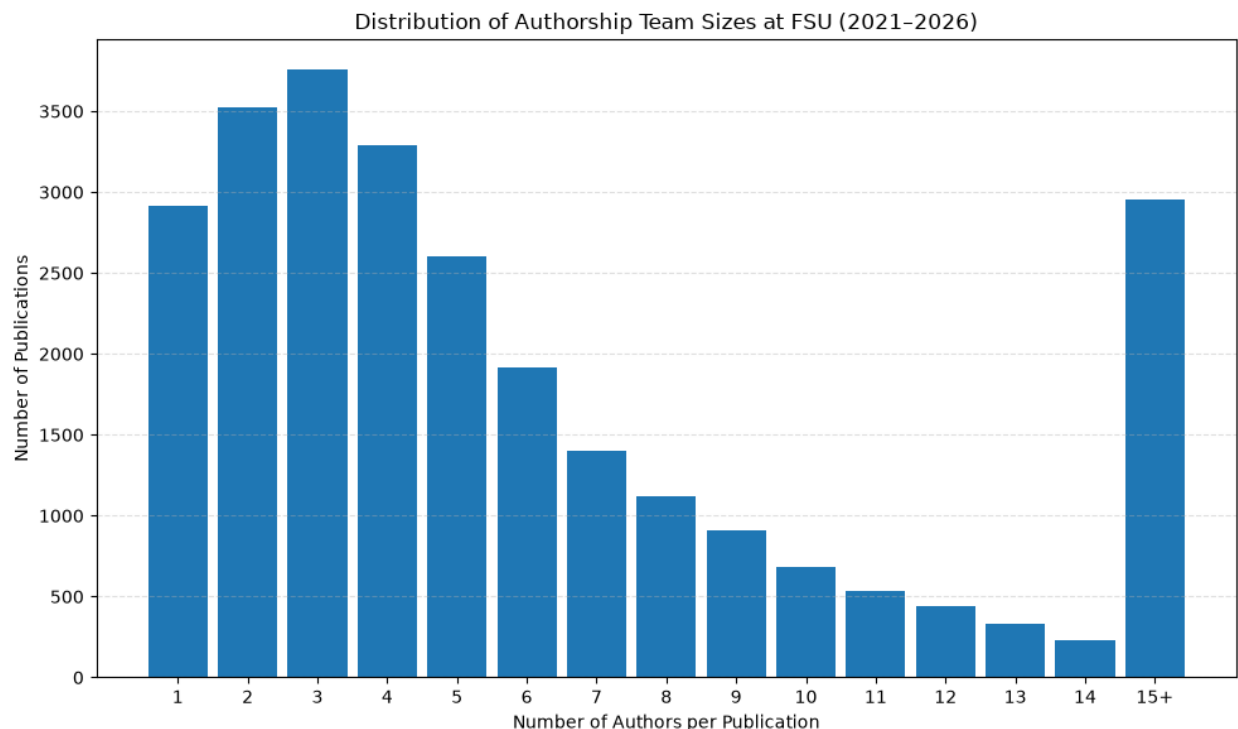


Chart 3 shows that most publications have around 3 authors, however, it hides that maybe this varies a lot between fields.

Citation Network

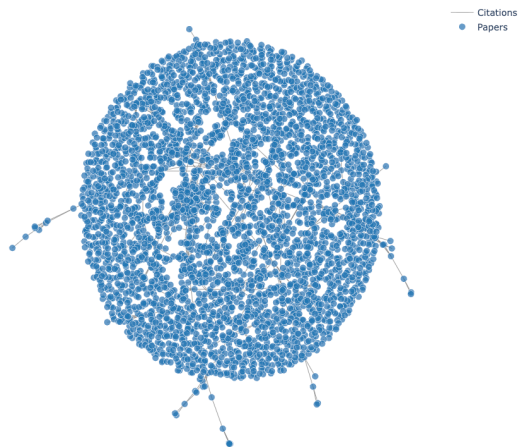
I picked a sample size 5000, as this was recommended with in the project instructions. It seemed to provide a satisfactory number of connections, but I opted not to do more as it would have taken a long time to download. This was sampled using a random sample of the fsu_works csv. This was unseeded.

The result was

```
... Fetching citations: 60% [██████████] | 2988/5000 [23:25<52:41:55, 94.29s/it]
Failed to fetch W4409087083: ('Connection aborted.', ConnectionResetError(54, 'Connection reset by peer'))
Fetching citations: 100% [██████████] | 5000/5000 [44:56<00:00, 1.85it/s]
```

From the graphs below, I can explore the connections between papers. However, a key limitation is I cannot observe the connections outside of FSU.

FSU-Internal Citation Network



The network drawn with NetworkX positions and Plotly traces

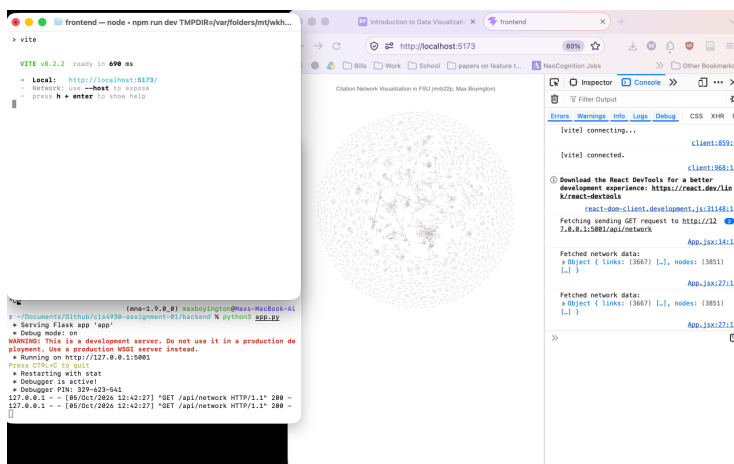
FSU-Internal Citation Network



The network after zooming into a region

Interactive D3 Visualization

Image of the graph



Demonstration that the frontend and backend communicate

Network Analysis

Q1: How large is the citation network?

Total Unique Papers (Nodes): 3,851

Total Unique Directed Citation Links (Edges): 3,667

Q2: What are the 10 most cited papers?

Top 10 Most Cited Papers by OpenAlex Global Citations:

Title	Year	Venue	Citation Count
Review of Particle Physics	2022	Progress of Theoretical and Experimental Physics	6344
Review of Particle Physics	2024	Physical review. D/Physical review. D.	3544
AmberTools	2023	Journal of Chemical Information and Modeling	2145
Fly Cell Atlas: A single-nucleus transcriptomi...	2022	Science	899
Increases in depression and anxiety symptoms i...	2021	Psychological Medicine	787
Polygenic prediction of educational attainment...	2022	Nature Genetics	715
Engagement in language learning: A systematic ...	2021	Language Teaching Research	658
A somato-cognitive action network alternates w...	2023	Nature	649
Ultrahard magnetism from mixed-valence dilanth...	2022	Science	614
Mesenchymal Stem Cell-Derived Exosomes: Applic...	2021	Cells	613